MATH 152 **QUIZ 5** NAME Solutions

Mrs. Bonny Tighe 25 points

7.8,8.1,8.2 TA_____ Wed. 3/13/2024

1. The area under a curve can be found if the improper definite integral is Convergent

2. Describe the two types of improper integrals.

1- where a limit of integration is $\pm\infty$

2- where the integral is discontinuous at a limit of integration

3. State the Comparison Test for Improper Integrals Suppose f and g are continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$

a) If $\int_a^\infty f(x)dx$ is convergent then $\int_a^\infty g(x)dx$ is also convergent

b) If $\int_a^\infty g(x)dx$ is divergent then $\int_a^\infty f(x)dx$ is also divergent

4. Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

a) $\int_2^5 \frac{1}{x^2-4}\,dx$

$\frac{A}{x+2} + \frac{B}{x-2} = \frac{1}{(x+2)(x-2)}$

$A(x-2) + B(x+2) = 1$

$x=2$ $B = 1/4$

$x=-2$ $A = -1/4$

$\lim_{t\to 2^+} \int_t^5 \left(\frac{-1/4}{x+2} + \frac{1/4}{x-2}\right)dx$

$\lim_{t\to 2^+} -\frac{1}{4}\ln|x+2| + \frac{1}{4}\ln|x-2| \Big]_t^5$

$\lim_{t\to 2^+} \frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| \Big]_t^5$

$\lim_{t\to 2^+} \frac{1}{4}\ln\left|\frac{5-2}{5+2}\right| - \frac{1}{4}\ln\left|\frac{0}{2}\right|$

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So $\int_2^5 \frac{1}{x^2-4}dx$ is also divergent

b) $\int_1^\infty \frac{1}{1+x^2}dx$

$\lim_{t\to\infty} \int_1^t \left(\frac{1}{1+x^2}\right)dx$

$\lim_{t\to\infty} \tan^{-1}x \Big|_1^t$

$\lim_{t\to\infty} \left(\tan^{-1}t - \tan^{-1}(1)\right)$

$\pi/2 - \pi/4 = \pi/4$

So $\int_1^\infty \left(\frac{1}{1+x^2}\right)dx$ is convergent

5. Use the comparison test to determine whether the integrals diverge or converge.

$$\int_2^\infty \frac{3x^2}{\sqrt{x^5-4x-3}}dx$$

Compare to $\int_2^\infty \frac{1}{\sqrt{x}}dx$ which is divergent $p = 1/2 \le 1$.

$$\frac{1}{\sqrt{x}} \le \frac{3x^2}{\sqrt{x^5-4x-3}}$$

So $\int_2^\infty \frac{3x^2}{\sqrt{x^5-4x-3}}dx$ is also divergent

$$\sqrt{x^5-4x-3} \le 3x^2\sqrt{x}$$

$$x^5-4x-3 \le 9x^5$$

$0 \le 8x^5+4x+3$ which is true for all $x \ge 2$

6. Find the length of the arc of the parabola $y = \frac{x^3}{6} + \frac{1}{2x}$, $\frac{1}{2} \le x \le 1$

$$L = \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$$

$$y = \frac{1}{6}x^3 + \frac{1}{2}x^{-1}$$

$$dy/dx = \frac{1}{2}x^2 - \frac{1}{2}x^{-2} = \frac{1}{2}x - \frac{1}{2x^2}$$

$$(dy/dx)^2 = \frac{1}{4}x^4 - \frac{1}{2} + \frac{1}{4x^4}$$

$$L = \int_{1/2}^1 \sqrt{1+\frac{1}{4}x^4 - \frac{1}{2} + \frac{1}{4x^4}} = \int_{1/2}^1 \sqrt{\frac{1}{4}x^4 + \frac{1}{2} + \frac{1}{4x^4}}\,dx$$

$$L = \int_{1/2}^1 \sqrt{\left(\frac{1}{2}x^2 + \frac{1}{2x^2}\right)^2}\,dx = \int_{1/2}^1 \left(\frac{1}{2}x^2 + \frac{1}{2x^2}\right)dx$$

$$L = \int_{1/2}^1 \left(\frac{1}{2}x^2 + \frac{1}{2}x^{-2}\right)dx = \frac{1}{2}\cdot\frac{1}{3}x^3 + \frac{1}{2}\cdot\frac{1}{-1}x^{-1}\Big]_{1/2}^1$$

$$\frac{1}{6}x^3 - \frac{1}{2x}\Big]_{1/2}^1 = \left(\frac{1}{6}-\frac{1}{2}\right) - \left(\frac{1}{6}\left(\frac{1}{8}\right) - 1\right)$$

$$\frac{1}{6} - \frac{1}{2} - \frac{1}{48} + 1 = 1/2 + 1/6 - 1/48$$

7. Set up an integral, **but do not solve**, for the area of the surface obtained by rotating the curve $y = \ln x$, $1 \le x \le 3$ about a) the x-axis and b) the y-axis

a) $2\pi \int_1^3 \ln x \sqrt{1 + (1/x)^2}\, dx$

b) $2\pi \int_1^3 x \sqrt{1 + (1/x)^2}\, dx$

8. The given curve is rotated about the x-axis. Find the area of the resulting surface.

$y = \sqrt{1+e^x}$, $0 \le x \le 1$ $\qquad S = 2\pi \int_a^b radius\,(arclength)$

$dy/dx = \frac{1}{2}(1+e^x)^{-1/2} e^x$

$dy/dx = \frac{e^x}{2(1+e^x)^{1/2}}$

$(dy/dx)^2 = \frac{e^{2x}}{4(1+e^x)}$

$2\pi \int_0^1 \sqrt{1+e^x} \sqrt{1 + \frac{e^{2x}}{4(1+e^x)}}\, dx$

$2\pi \int_0^1 \sqrt{1+e^x} \sqrt{\frac{4+4e^x+e^{2x}}{4(1+e^x)}}\, dx$

$2\pi \int_0^1 \frac{\cancel{\sqrt{1+e^x}}\, \sqrt{(e^x+2)^2}}{2\cancel{\sqrt{1+e^x}}}\, dx$

$2\pi \int_0^1 \frac{1}{2}(e^x+2)\, dx$

$\pi \left[e^x + 2x \right]_0^1 =$

$\pi \left[(e+2) - (e^0+0) \right] = \pi(e+1)$

MATH 152
Mrs. Bonny Tighe

QUIZ 5A
25 points
7.8,8.1,8.2

NAME Solutions

TA_____ Wed. 3/13/2024

(2pts)

1. State the Comparison Test for Improper Integrals

Suppose that f and g are continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$ a) If $\int_a^\infty f(x)dx$ is convergent then $\int_a^\infty g(x)dx$ is too

b) If $\int_a^\infty g(x)dx$ is divergent, then $\int_a^\infty f(x)dx$ is divergent too

2. The area under a curve can be found if the improper definite integral is Convergent

(2pts)

3. Describe the two types of improper integrals.

1- where a limit of integration is $\pm\infty$

2- where the integrand is discontinuous at a limit of integration

4. Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

a) $\int_1^2 \frac{1}{x\sqrt{x^2-1}}\,dx$ (3pts)

$\lim_{t\to 1^+} \int_t^2 \frac{1}{x\sqrt{x^2-1}}\,dx$

$\lim_{t\to 1^+} \sec^{-1}x \Big|_t^2$

$\lim_{t\to 1^+} \left(\sec^{-1}(2) - \sec^{-1}(t)\right)$

$\pi/3 - 0$

$\pi/3$

So $\int_1^2 \frac{1}{x\sqrt{x^2-1}}\,dx$ is Convergent

(3pts) b) $\int_0^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}}\,dx$

$u = -\sqrt{x}$

$du = -\frac{1}{2\sqrt{x}}\,dx$

$(-2)\lim_{t\to\infty} \int_0^t e^{\sqrt{x}}\,du$

$-2\lim_{t\to\infty} e^{-\sqrt{x}}\Big|_0^t$

$-2\lim_{t\to\infty}\left[\frac{1}{e^{\sqrt{t}}} - \frac{1}{e^0}\right]$

$-2[0-1] = 2$

So $\int_0^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}}\,dx$ is Convergent

$$\frac{x^3}{x^{7/2}}$$

5. Use the comparison test to determine whether the integrals diverge or converge.

$$\int_2^\infty \frac{x^3}{\sqrt{x^7-4x-3}}dx$$

Compare to $\int_2^\infty \frac{1}{x^{1/2}}dx$ which is a divergent indefinite integral $p = 1/2 \le 1$

4pts

$$\frac{1}{\sqrt{x}} \le \frac{x^3}{\sqrt{x^7-4x-3}}$$

$$\sqrt{x^7-4x-3} \le x^3\sqrt{x}$$

$$x^7-4x-3 \le x^7$$

$$0 \le 4x+3$$

which is true for all $x \ge 2$

So $\int_2^\infty \frac{x^3}{\sqrt{x^7-4x-3}}dx$ is also divergent

6. Find the length of the arc $x = \frac{y^4}{8} + \frac{1}{4y^2}$ $1 \le y \le 2$ $L = \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$

4pts

$$x = \frac{1}{8}y^4 + \frac{1}{4}y^{-2}$$

$$dx/dt = \frac{1}{2}y^3 - \frac{1}{2}y^{-3}$$

$$= \frac{1}{2}y^3 - \frac{1}{2y^3}$$

$$(dx/dt)^2 = \frac{1}{4}y^6 - \frac{1}{2} + \frac{1}{4y^6}$$

$$L = \int_1^2 \sqrt{\frac{1}{4}y^6 - \frac{1}{2} + \frac{1}{4y^6} + 1}\,dy$$

$$L = \int_1^2 \sqrt{\frac{1}{4}y^6 + \frac{1}{2} + \frac{1}{4y^6}}\,dy$$

$$L = \int_1^2 \sqrt{\left(\frac{1}{2}y^3 + \frac{1}{2y^3}\right)^2}\,dy$$

$$L = \int_1^2 \left(\frac{1}{2}y^3 + \frac{1}{2y^3}\right)dy$$

$$\frac{1}{2}\cdot\frac{1}{4}y^4 + \frac{1}{2}\,\frac{1}{-2}y^{-2}\Big]_1^2$$

$$\frac{1}{8}y^4 - \frac{1}{4y^2}\Big]_1^2 = \frac{1}{8}(16) - \frac{1}{4(1)} = 2 - \frac{1}{4} = 7/4$$

2pts 7. Set up an integral, **but do not solve,** for the area of the surface obtained by rotating the curve $y = e^x$, $1 \le x \le 3$ about a) the x-axis and b) the y-axis

a) $2\pi \int_1^3 e^x \sqrt{1+e^{2x}}\, dx$

b) $2\pi \int_1^3 x \sqrt{1+e^{2x}}\, dx$

8. The given curve is rotated about the x-axis. Find the area of the resulting surface.

$r = y$

4pts $y = \sqrt{1+e^x}$, $0 \le x \le 1$ $\quad S = 2\pi \int_a^b radius(arclength))$

$dy/dx = \frac{1}{2}(1+e^x)^{-1/2} e^x$

$dy/dx = \frac{e^x}{2\sqrt{1+e^x}}$

$(dy/dx)^2 = \frac{e^{2x}}{4(1+e^x)}$

$S = 2\pi \int_0^1 \sqrt{1+e^x} \sqrt{1 + \frac{e^{2x}}{4(1+e^x)}}\, dx$

$S = 2\pi \int_0^1 \sqrt{1+e^x} \sqrt{\frac{4+4e^x+e^{2x}}{4(1+e^x)}}\, dx$

$S = 2\pi \int_0^1 \cancel{\sqrt{1+e^x}} \frac{\sqrt{e^{2x}+4e^x+4}}{2\cancel{\sqrt{1+e^x}}}\, dx$

$S = 2\pi \int_0^1 \frac{\sqrt{(e^x+2)^2}}{2}\, dx = \pi \int_0^1 (e^x+2)\, dx$

$\pi \left[e^x + 2x\right]_0^1 = \pi \left[(e^1+2) - (e^0+0)\right]$

$\pi [e+1]$